# Our Adaptation Tutorial

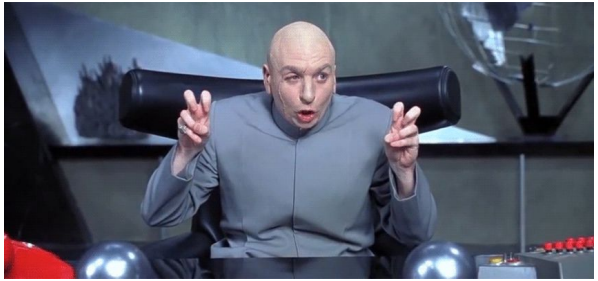
Larry & Caleb

# How not to do adaptation:

- Just guess
- Try to be clever with data
- “My partner is the expert”
- Dropping the class
- Hoping your partner has an epiphany

# What works:

- Try a small example
- Basic rules-of-thumb from your local “expert”



2012 Mercedes-Benz E-Class
(BLACK)

2014 Mercedes-Benz E-Class
(GRAY)

Retrieved Case:

```
make_name: Mercedes-Benz
model_name: E-Class
year: 2014

body_type: Sedan
maximum_seating: 5

city_fuel_economy: 20.0
daysonmarket: 24
engine_cylinders: V6
engine_displacement: 3500.0
fleet: True
frame_damaged: True
fuel_tank_volume: 21.1
fuel_type: Gasoline
has_accidents: True
highway_fuel_economy: 27.0
horsepower: 302.0
listing_color: GRAY
mileage: 76930.0
owner_count: 2
transmission: A
wheel_system: AWD

price: 18825.0
```

Problem Case:

```
make_name: Mercedes-Benz
model_name: E-Class
year: 2012

body_type: Sedan
maximum_seating: 5

city_fuel_economy: 20.0
daysonmarket: 86
engine_cylinders: V6
engine_displacement: 3500.0
fleet: True
frame_damaged: True
fuel_tank_volume: 21.1
fuel_type: Gasoline
has_accidents: True
highway_fuel_economy: 30.0
horsepower: 302.0
listing_color: BLACK
mileage: 99254.0
owner_count: 3
transmission: A
wheel_system: RWD

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        mileage: 99254.0

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```

Is this problem car cheaper?  More expensive?

```
year:          ↓
daysonmarket:  ↑
price:         ↑
```

```
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        year: 2014
        daysonmarket: 24
        mileage: 76930.0

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```

```
Problem Case:
        year: 2012
        daysonmarket: 86
        mileage: 99254.0

        price: ???
```

By how much?

```
year:
daysonmarket:
price:
```

We can cheat and peek at the actual price:
$12492

# Rules of Thumb

1. Every year, the expected price of a car increases by 10%
2. Every 90 days on the lot, the expected price drops by 0.5%
3. Every 10,000 miles, the expected price depreciates by 4%

$18825 - ($3765.0 + $64.9 + $1681.1) = $13314

Compare to actual: $12492